

PORT WASHINGTON, WI, USA

WMO: 997736

Lat: 43.3881N

Lon: 87.8667W

Elev: 180

StdP: 99.18

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.5	-16.8	N/A	N/A	N/A	N/A	N/A	N/A	11.7	-2.3	10.7	-1.0	4.4	280	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	30.5	N/A	28.3	N/A	26.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.1	250

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Externe Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	8.4	7.4	-21.8	34.7	2.7	2.4	-23.8	36.4	-25.4	37.9	-26.9	39.2	-28.9	41.0		
			DB	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-5.8	-5.1	0.8	6.4	12.0	17.4	21.2	21.0	17.1	10.5	3.9	-1.8
	DBStd	10.41	6.46	5.72	5.06	3.57	3.51	3.75	3.45	2.91	3.49	3.88	4.95	5.37
	HDD10.0	1932	491	422	287	118	18	0	0	0	1	42	189	364
	HDD18.3	3947	749	656	543	357	198	62	12	9	63	242	433	623
	CDD10.0	1281	0	0	1	11	79	223	346	342	213	59	7	0
	CDD18.3	256	0	0	0	0	2	34	100	93	26	1	0	0
	CDH23.3	1359	0	0	0	4	12	206	612	417	102	7	0	0
	CDH26.7	377	0	0	0	1	1	50	203	97	23	1	0	0
Wind	WSAvg	3.3	3.9	4.0	3.8	3.9	3.1	2.6	2.5	2.6	2.9	3.4	3.6	3.7
Precipitation	PrecAvg	855	44	36	49	88	98	116	96	94	68	61	53	44
	PrecMax	1076	118	80	113	158	297	336	192	209	176	136	134	99
	PrecMin	699	8	3	9	23	25	37	48	51	0	18	8	8
	PrecStd	100	25	22	28	36	59	72	42	41	38	35	35	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	10.0	16.4	21.3	26.9	30.7	35.2	32.5	29.5	23.8	17.8	11.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	6.7	5.3	12.4	16.6	22.6	28.1	31.3	29.4	25.8	19.5	14.2	8.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	4.6	3.5	10.0	13.5	20.0	25.8	28.9	26.9	23.9	17.8	11.6	6.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	3.0	2.2	7.5	11.2	17.5	23.3	26.6	25.4	22.2	16.2	10.1	5.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.1	6.8	6.4	7.2	7.3	7.8	7.4	7.3	7.1	6.5	6.4
		MCDBR	8.5	7.9	9.9	10.6	11.5	11.6	11.3	10.0	9.7	9.6	8.2	7.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.288	0.316	0.342	0.389	0.412	0.431	0.424	0.417	0.390	0.346	0.309	0.287	
	taud	2.429	2.336	2.374	2.284	2.274	2.269	2.292	2.305	2.365	2.476	2.518	2.475	
	Ebn at Noon	850	877	897	875	862	843	845	839	835	835	821	819	
	Edh at Noon	74	97	107	126	131	133	128	122	107	84	69	65	
All-Sky Solar Radiation	RadAvg	1.48	2.42	3.60	4.56	5.36	6.01	6.16	5.27	4.25	2.70	1.68	1.18	
	RadStd	0.20	0.27	0.36	0.42	0.44	0.42	0.28	0.33	0.35	0.31	0.15	0.12	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page